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**P113 – A2LA Policy on Measurement Traceability for
Life Sciences Testing Laboratories**

(November 2012)

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Scope and Field of Application:

This document serves as the primary measurement traceability policy for “Life Sciences” testing laboratories (replacing *P102 – A2LA Policy on Measurement Traceability*). It is meant to explain how measurement traceability may be achieved and how it may be demonstrated for biological, environmental and chemical testing laboratories. This document is meant to be used in conjunction with *P102a – Policy on Reference Material Traceability for Life Sciences Testing Laboratories*, which explains how biological (including organisms) and chemical materials (including drugs) are expected to be processed and how they may be used within life sciences laboratories to meet traceability requirements.

NOTE: *P102b – Policy on Requesting an Exception to Measurement Traceability* outlines the requirements to request and receive an exception to this traceability policy.

Introduction and Discussion:

The following premises and statements are fundamental to implementing this Traceability Policy. Appendix I contains definitions of terms used throughout this document. An awareness of these definitions is essential to understanding and implementing this Policy.

Life Science Laboratories are expected to demonstrate metrological traceability between the measurement results from their equipment and measurement standards. This multi-step process may be thought of as involving, as the first step, solving a relationship between the measured values and intermediate variables, and as a subsequent step solving an algebraic relationship between these intermediate variables and the error source. It is inferred that:

- Calibration of equipment can be performed using measurement standards that include both reference standards and reference materials.
- There is no fundamental difference between calibrating balances or gas chromatographs, with respect to metrological traceability.
- When ISO/IEC 17025:2005, Section 5.6.3.1 refers to “reference standards” it is referring to both “measurement standards” (referenced in VIM 5.1) and “reference measurement standards” or “reference standards” (referenced in VIM 5.6), such that:
 - Phrases like the “reference standards shall be calibrated before and after any adjustment” in Section 5.6.3.1 of ISO/IEC 17025:2005 would apply to both types of “standards” mentioned above.
 - The use of intermediate checks are used to check the “calibration status of reference, primary, transfer or working standards” (those identified in VIM 5.4-5.8), as emphasized in ISO/IEC 17025:2005, Section 5.6.3.3.
- With respect to reference materials, VIM 5.13 introduces three new terms: “material”, “homogeneous” and “stable” (none of which are defined in the VIM). A2LA accepts the following definition associated with these terms in the following ISO Guide 35 definition:
 - Reference Material (3.1): material sufficiently homogeneous and stable with respect to one or more specified properties, which has been established to be fit for its intended use in a measurement process. (ISO Guide 35)

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Also with respect to reference materials, in Figure A.12 of the VIM, reference standards all appear on the left side of the diagram, while the reference materials all appear on the right side, which appears to categorize them somewhat differently.

(LST-1) A2LA requires that:

(a) All calibrations of reference standards (i.e. reference masses and reference gauge blocks) and measuring equipment (i.e. reference thermometers, balances and pipettors) shall be conducted by:

- A calibration laboratory accredited to ISO/IEC 17025 by a mutually recognized Accreditation Body; or,
- A recognized National Metrology Institute (NMI), including designated institutes. Recognition of the NMI is based on the Institute being a signatory to the CIPM (Comité International des Poids et Mesures) MRA (Mutual Recognition Arrangement) and supporting the measurement comparison activities of the CIPM. A listing of these recognized Institutes may be found at <http://kcdb.bipm.org/default.asp>; or,
- The testing laboratory itself, if it is accredited by A2LA to ISO/IEC 17025 and found to meet the LST-4 requirements of this document for their in-house calibrations; or,
- A State Weights and Measures facility with a current certificate of measurement traceability. Please see <http://ts.nist.gov/WeightsAndMeasures/statelabcontact.cfm> for a copy of current certificates.

(b) All calibrations (i.e., Initial Calibration or ICAL - once known as standardization) of test equipment (i.e. GC, ICP, HPLC, OES, AA, LECO, SEM, UV/VIS) shall be conducted using Reference Materials (when they are relevant and available) obtained from:

- A reference material producer accredited to ISO Guide 34 in combination with ISO/IEC 17025 by an Asia Pacific Laboratory Accreditation Cooperation (APLAC) Mutual Recognition Arrangement (MRA) signatory that is recognized for accrediting reference material producers; or,
- A recognized National Metrology Institute (NMI) or designated institute; or,
- An organization meeting the requirements of P102a - Policy on Reference Material Traceability for Life Sciences Testing Laboratories.

LST-1(b) is required unless it is shown that the laboratory's customer does not require traceability to the SI. The fact that the customer does not require the reporting of measurement uncertainty on the final report SHALL be confirmed in writing and recorded during contract review (ISO/IEC 17025, section 4.4, unless it contradicts the requirements of 5.10.3.1.c). In these cases, the reference material may be obtained from a reference material producer meeting the purchasing criteria of the laboratory. The purchasing criteria of the laboratory shall ensure that the reference material is appropriate for the intended use (i.e. a reference method requirement defined by regulation, government organization or customer requirement).

(NOTE 1: Refer to Appendix II for details on required calibrations and intermediate checks of M&TE when traceability to the SI is required by the customer).

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(LST-2) *For external calibrations of reference standards and measuring equipment requiring traceability through reference standards, these shall be recorded in a calibration certificate or report endorsed by the recognized Accreditation Body's symbol (or otherwise makes reference to accredited status by a specific, recognized accreditation body), the measurement uncertainty and an indication of the type of entity that is accredited (e.g. inclusion of "calibration laboratory" with the symbol, etc.) or as endorsed by a National Metrology Institute (NMI). A2LA-accredited organizations are also required to include their A2LA certificate number along with an indication of the type of entity with every use of the A2LA Accredited symbol (or narrative reference) (see A2LA P101, Section B2.2). For internal calibrations, the requirements outlined in requirement LST-4 of this document shall apply.*

(LST-3) *All A2LA-accredited and enrolled organizations shall define their policy for achieving measurement traceability and also for achieving traceability for reference materials, when applicable. The policy shall ensure compliance with this policy document.*

(LST-4) *All in-house calibrations of reference standards and measuring equipment requiring traceability through reference standards (i.e. reference masses, reference thermometers, reference gauge blocks, balances, pipettors) shall be supported by the following minimal set of elements:*

- a) The in-house laboratory shall maintain documented procedures for the in-house calibrations and the in-house calibrations shall be evidenced by a calibration report, certificate, sticker, or other suitable method, and calibration records shall be retained for an appropriate, prescribed time;*
- b) The in-house laboratory shall maintain training records for calibration personnel and these records shall demonstrate the technical competence of the personnel performing the calibrations. Evidence of competence SHALL include measurement results, for example, recorded training and the results of measurement audits;*
- c) The in-house laboratory shall be able to demonstrate traceability to national or international standards of measurement by procuring calibration services from appropriately accredited calibration laboratories or an NMI;*
- d) The in-house laboratory shall have and apply procedures for evaluating measurement uncertainty. Measurement uncertainty shall be calculated for each type of calibration and records of these estimates shall be maintained. Measurement uncertainty shall be taken into account when statements of compliance with specifications are made; and,*
- e) Reference standards shall be recalibrated at appropriate intervals to ensure that the reference value is reliable. Policy and procedures for establishing and changing calibration intervals shall be based on the historical behavior of the reference standard.*

(LST-5) *Required in-house calibrations of test equipment requiring traceability through reference materials (i.e. gas chromatographs, atomic absorption spectrophotometers and pH meters) shall be supported by the following minimal set of elements:*

- a) The in-house laboratory shall maintain documented procedures for the in-house calibrations of measuring and test equipment and the in-house calibrations shall be evidenced by calibration records and the calibration records shall be retained for an appropriate, prescribed time;*

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- b) *The in-house laboratory shall maintain training records for calibration personnel and these records shall demonstrate the technical competence of the personnel performing the calibrations. Evidence of competence SHALL include measurement results, for example, recorded training and the results of proficiency or quality control testing results;***
- c) *The in-house laboratory shall be able to demonstrate traceability to reference materials from accredited reference material producers or as defined in (LST-1 b) above; and,***
- d) *The in-house laboratory shall have and apply procedures for evaluating method measurement uncertainty. Note: the method measurement uncertainty may be based on the entire process such as the laboratory control sample, or may be based on each step of the analysis process. Measurement uncertainty estimates shall be calculated and records of these estimates shall be maintained. The estimation of uncertainty may be accomplished with the effective use of and analysis of control charts, control limits and other processes. See A2LA P103b - Measurement Uncertainty for Life Science Laboratories.***

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APPENDIX I

Definition of Terms:

➤ **VIM 2.41: metrological traceability -**

Property of a measurement result whereby the result can be related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty.

VIM Note 7 - ILAC considers the elements for confirming metrological traceability to be an unbroken metrological traceability chain to an international measurement standard or a national measurement standard, a documented measurement uncertainty, a documented measurement procedure, accredited technical competence, metrological traceability to the SI, and calibration intervals (see ILAC P-10).

The following definitions relate to the phrase “measurement result”, as found in the VIM 2.41 definition of “metrological traceability”:

➤ **VIM 2.9: measurement result; result of measurement -**

Set of quantity values being attributed to a measurand together with any other available relevant information.

VIM Note 2 - A measurement result is generally expressed as a single measured quantity value and a measurement uncertainty. If the measurement uncertainty is considered to be negligible for some purpose, the measurement result may be expressed as a single measured quantity value. In many fields, this is the common way of expressing a measurement result.

A2LA Practical Example - 1: 25 µg/L of benzene in water

A2LA Practical Example - 2: 0.5 g/L of lead in 1 dL of blood

➤ **VIM 1.1: quantity -**

Property of a phenomenon, body, or substance where the property has a magnitude that can be expressed as a number and a reference.

VIM Note 1 Example - Amount-of-substance concentration in entity B, c_B : amount-of-substance concentration of ethanol in wine sample i , c_i (C_2H_5OH).

VIM Note 4 Example - The preferred IUPAC-IFCC format for designations of quantities in laboratory medicine is “System—Component; kind-of-quantity”: “Plasma (Blood)—Sodium ion; amount-of-substance concentration equal to 143 mmol/l in a given person at a given time”.

➤ **VIM 1.19: quantity value; value of a quantity; value -**

Number and reference together expressing magnitude of a quantity.

VIM Example 1 - Length of a given rod: 5.34 m or 534 cm

VIM Example 2 - Mass of a given body: 0.152 kg or 152 g

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➤ **VIM 2.3: measurand -**

Quantity intended to be measured.

VIM Note 1 - The specification of a measurand requires knowledge of the kind of quantity, description of the state of the phenomenon, body, or substance carrying the quantity, including any relevant component, and the chemical entities involved.

VIM Note 4 - In chemistry, “analyte”, or the name of a substance or compound, are terms sometimes used for ‘measurand’. This usage is erroneous because these terms do not refer to quantities.

A2LA Practical Example 1 - Mass of benzene in water expressed in µg/L

A2LA Practical Example 2 - Mass of lead in a given volume of blood expressed in g/L

The following definitions relate to the term “reference”, as found in the VIM 2.41 definition of “metrological traceability”:

➤ **VIM 5.1: measurement standard -**

Realization of the definition of a given quantity, with stated quantity value and associated measurement uncertainty, used as a reference.

VIM Example 1 - 1 kg mass measurement standard with an associated standard measurement uncertainty of 3 µg.

VIM Example 5 - Set of reference solutions of cortisol in human serum having a certified quantity value with measurement uncertainty for each solution.

VIM Example 6 - Reference material providing quantity values with measurement uncertainties for the mass concentration of each of ten different proteins.

➤ **VIM 5.6: reference measurement standard or reference standard -**

Measurement standard designated for the calibration of other measurement standards for quantities of a given kind in a given organization or at a given location.

A2LA Practical Example 1 - Calibrating a working standard mass with a reference standard mass.

A2LA Practical Example 2 - Calibrating a working thermometer with a NIST-traceable reference thermometer.

➤ **VIM 5.7 working measurement standard; working standard -**

Measurement standard that is used routinely to calibrate or verify measuring instruments or measuring systems.

VIM Note 1 - A working measurement standard is usually calibrated with respect to a reference measurement standard.

VIM Note 2 - In relation to verification, the terms “check standard” or “control standard” are also sometimes

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used.

➤ **VIM 5.13: reference material; RM -**

Material, sufficiently homogeneous and stable with reference to specified properties, which has been established to be fit for its intended use in measurement or in examination of nominal properties.

VIM Note 1 - Examination of a nominal property provides a nominal property value and associated uncertainty. This uncertainty is not a measurement uncertainty.

VIM Note 2 - Reference materials with or without assigned quantity values can be used for measurement precision control whereas only reference materials with assigned quantity values can be used for calibration or measurement trueness control.

VIM Note 3 - ‘Reference material’ comprises materials embodying quantities as well as nominal properties.

VIM Example 1 - Examples of reference materials embodying quantities:

- a) water of stated purity, the dynamic viscosity of which is used to calibrate viscometers;
- b) human serum without an assigned quantity value for the amount-of-substance concentration of the inherent cholesterol, used only as a measurement precision control material;
- c) fish tissue containing a stated mass fraction of a dioxin, used as a calibrator.

VIM Example 2 - Examples of reference materials embodying nominal properties:

- a) colour chart indicating one or more specified colours;
- b) DNA compound containing a specified nucleotide sequence;
- c) urine containing 19-androstenedione.

➤ **VIM 5.14 (6.14) certified reference material; CRM -**

reference material, accompanied by documentation issued by an authoritative body and providing one or more specified property values with associated uncertainties and traceabilities, using valid procedures.

The following definitions relate to the term “calibration”, as found in the VIM 2.41 definition of “metrological traceability”:

➤ **VIM 2.39: calibration**

Operation that, under specified conditions, in a first step, establishes a relation between the quantity values with measurement uncertainties provided by measurement standards and corresponding indications with associated measurement uncertainties and, in a second step, uses this information to establish a relation for obtaining a measurement result from an indication.

VIM Note 1 - A calibration may be expressed by a statement, calibration function, calibration diagram, calibration curve, or calibration table. In some cases, it may consist of an additive or multiplicative correction of the indication with associated measurement uncertainty.

VIM Note 2 - Calibration should not be confused with adjustment of a measuring system, often mistakenly called “self-calibration,” nor with verification of calibration.

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VIM Note 3 - Often, the first step alone in the above definition is perceived as being calibration.

➤ **VIM 2.44: verification -**

Provision of objective evidence that a given item fulfills specified requirements.

VIM Example 1 - Confirmation that a given reference material as claimed is homogeneous for the quantity value and measurement procedure concerned, down to a measurement portion having a mass of 10 mg.

VIM Example 2 - Confirmation that performance properties or legal requirements of a measuring system are achieved.

VIM Note 1 - When applicable, measurement uncertainty should be taken into consideration.

VIM Note 2: The item may be, e.g., a process, measurement procedure, material, compound, or measuring system.

VIM Note 3: The specified requirements may be, e.g., that a manufacturer's specifications are met.

VIM Note 5: Verification should not be confused with calibration. Not every verification is a validation.

A2LA Practical Example 1: Verification = Intermediate Check (ISO/IEC 17025:2005 Section 5.5.10)

A2LA Practical Example 2: Continuing Calibration Verification (CCV or ICV) in environmental testing is a “verification”

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APPENDIX II

Equipment Calibration and Verification When Traceability to the SI is a Customer Requirement

NOTE: If not otherwise specified (i.e. by the method as defined in ISO/IEC 17025 5.4.2 or client requirements), these are the minimum actions required unless the CAB can show that the uncertainty contribution of the specified equipment is minimal to the outcome of the measurement (i.e. $<1/3$ of the expanded uncertainty U [VIM 2.35]). Please see references at end of table.

Also per ISO/IEC 17025 Section 5.5.2 there may be equipment not listed in this table that is deemed critical by the CAB and which therefore requires full traceable calibration.

	Minimum Action	Frequency*	Minimum Traceability
Reference Standards			
Weights (Mass) Calibration of Support Equipment Not Listed on Scope	- Shall Have Initial and Ongoing LST-1 or LST-4 Traceable Calibration	- As Defined by The Laboratory	- Requires Traceability At Least To An NMI
	- Should Do Intermediate Check	- Intermediate Check Should Be Done Annually	
Thermometer Calibration of Support Equipment Not Listed on Scope	- Shall Have Initial and Ongoing LST-1 or LST-4 Traceable Calibration	- As Defined By The Laboratory	- Requires Traceability At Least To An NMI
	- Should Perform Intermediate Check At Ice-Point or Boiling Point or At Temperature of Use (Versus Reference Thermocouple)	- Intermediate Check Should Be Performed Annually or As Defined By The Laboratory	

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	Minimum Action	Frequency	Minimum Traceability
Reference Standards			
Thermocouple Calibration of Support Equipment Not Listed on Scope	- Shall Have Initial and Ongoing LST-1 or LST-4 Traceable Calibration	- As Defined By The Laboratory	- Requires Traceability At Least To An NMI
	- Should Perform Intermediate Check At Ice-Point or Boiling Point	- Intermediate Check Should Be Performed Annually or As Defined By The Laboratory	
Gauge Block Calibration of Support Equipment Not Listed on Scope	- Shall Have Initial and Ongoing LST-1 or LST-4 Traceable Calibration	-As Defined By The Laboratory	- Potential Significant Component Of Overall Uncertainty Of Measurand (i.e., Cigarette Length And Circumference)
	- Should Perform Intermediate Checks	- Intermediate Check Should Be Performed Monthly or As Defined By The Laboratory	
Magnification Standard (a.k.a. Stage Micrometer) Support Equipment Not Listed on Scope (Used for calibration checks of ocular micrometers and reticles for optical microscopy)	- Shall Have Initial LST-1 or LST-4 Traceable Calibration	- As Defined By The Laboratory	- Requires Traceability At Least To An NMI
		- Usually Only Initial Calibration	
Measuring Instruments - In Scope Of Accreditation			
pH Meter/Electrode Gas Chromatograph Liquid Chromatograph AA Spectrophotometer Inductively Coupled Plasma UV/VIS Spectrophotometer (ICAL)	- Response Comparison With Reference Materials (Standards) Traceable (To NIST, NMI, RMP if Possible) or Authoritative Source - Measurement Uncertainty Determined With LCS Data for the Method	- As Defined By The Laboratory	- Major Component Of Overall Uncertainty Of Measurand - Already Part Of Test Method Uncertainty
(CCV or ICV)	- Compare To Reference Material	N/A	N/A

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	Minimum Action	Frequency	Minimum Traceability
Support Equipment/Apparatus			
Balances	- Shall Have Initial and Ongoing LST-1 or LST-4 Traceable Calibration - Measurement of Mass is Within Lab's Acceptance Criteria	- As Defined By The Laboratory	- Can Be An Insignificant Component Of Overall Uncertainty Of Measurement - Already Part Of Test Method Uncertainty
	- Should Perform Intermediate Check With Working Weights	- Intermediate Check Should Be Performed on Day of Use	
Weights	- Compare To Reference Weights - Measurement Of Mass Is Within Acceptance Criteria	- As Defined By The Laboratory	- Insignificant Component Of Overall Uncertainty - Already Part Of Test Method Uncertainty
Pipettors and Other Volumetric Delivery Devices	- Shall Have Initial and Ongoing LST-1 or LST-4 Traceable Calibration - Measurement of Mass is Within Lab's Acceptance Criteria	- As Defined By The Laboratory	- Potentially Significant Component Of Overall Uncertainty Of Measurand - Already Part Of Test Method Uncertainty.
	- Verification With Weight of Water or Spectrophotometric Acceptable	- As Defined By The Laboratory	
pH Meter/Electrode	- Response Comparison With Non-Traceable Buffers	- Should Be Performed Before Each Use	- Small Component Of Overall Uncertainty Of Measurand - Already Part Of Test Method Uncertainty
Refrigerators, Freezers, Incubators, Water Baths, Ovens, Muffle Furnaces, etc.	- None Required, - Using Appropriate Working Thermometers or Working Thermocouples is Expected	N/A	- Insignificant Component Of Overall Uncertainty Of Measurand - Already Part Of Test Method Uncertainty

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	Minimum Action	Frequency	Minimum Traceability
Autoclaves	- None Required - Using Appropriate Working Thermometers or Working Thermocouples is Expected	N/A	- Insignificant Component Of Overall Uncertainty Of Measurand - Already Part Of Test Method Uncertainty
Safety Cabinets, Fume Hoods, Laminar Flow Hoods, Centrifuges	- None Required Centrifuges May Need To Be Calibrated If The Laboratory Makes Them “Quality Critical” (E.G. 2000 Rpm ± 100 Rpm)	N/A	- Insignificant Component Of Overall Uncertainty Of Measurand - Already Part Of Test Method Uncertainty
Thermometers	- Compare To Reference Thermometer - Measurement Of Temperature Is Within Acceptance Criteria Or Correction Factor Applied	- As Defined By The Laboratory	- Insignificant Component Of Overall Uncertainty - Already Part Of Test Method Uncertainty
Thermal cyclers / PCR system	- Defined By Manufacturer or Best Practices of the Laboratory	- Defined By Manufacturer or Best Practices of the Laboratory	- Proprietary
Plate Readers (Conv., PF)	- Defined By Manufacturer or Best Practices of the Laboratory	- Defined By Manufacturer or Best Practices of the Laboratory	- Proprietary
Genetic Sequencers	- Defined By Manufacturer or Best Practices of the Laboratory	- Defined By Manufacturer or Best Practices of the Laboratory	- Proprietary
Automated ELISA	- Defined By Manufacturer or Best Practices of the Laboratory	- Defined By Manufacturer or Best Practices of the Laboratory	- Proprietary

ISO/IEC 17025 requires that, unless it has been established that the associated contribution from the calibration contributes little to the total uncertainty of the test result, the requirements given for calibration laboratories apply for measuring and test equipment with measuring functions used. When this situation arises, the laboratory needs to ensure that the equipment used can provide the uncertainty of measurement needed. However, the extent to which these requirements shall be

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followed depends on the relative contribution of the calibration uncertainty to the total uncertainty. If calibration is the dominant factor, the requirements shall be strictly followed.

With respect to reference materials, in those cases where traceability of the measurement to SI units is not requested by the customer, not possible, or not relevant, the same requirements for traceability to primary standards (i.e. certified reference materials), agreed methods and/or consensus standards apply.

References:

Eurachem/CITAC guide: Quantifying Uncertainty in Analytical Measurement, Third edition, (2012) ISBN 978-0-948926-30-3. This document is available from <https://www.eurachem.org>.

***ILAC G24/OIML D10: Guidelines for the determination of calibration intervals of measuring instruments. This document is available at https://www.ilac.org/documents/ILAC_G24_2007.pdf.**

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